

SHENZHEN UNIVERSITY
INSTITUTE FOR ADVANCED STUDY

Differential Equations, Autumn 2026

Assignment 1, due at 1:00 PM on Wednesday, September 30, 2026

1.1

In Problem 13-15, write a differential equation that fits the physical description.

13. The rate of change of the population p of bacteria at time t is proportional to the population at time t .

Answer.

$$\frac{dp}{dt} = kp$$

14. The velocity at time t of a particle moving along a straight line is proportional to the fourth power of its position x .

Answer.

$$v = kx^4$$

1.2

- 1.(a) Show that $\phi(x) = x^2$ is an explicit solution to

$$x \frac{dy}{dx} = 2y$$

on the interval $(-\infty, \infty)$.

Answer.

$$\phi(x) = x^2, \quad x \in (-\infty, \infty)$$

$$x \cdot \frac{dy}{dx} = x \cdot (x^2)' = x \cdot 2x = 2 \cdot x^2 = 2\phi(x) = 2y$$

(b) Show that $\phi(x) = e^x - x$ is an explicit solution to

$$\frac{dy}{dx} + y^2 = e^{2x} + (1 - 2x)e^x + x^2 - 1$$

on the interval $(-\infty, \infty)$.

Answer.

$$\phi(x) = e^x - x, \quad x \in (-\infty, \infty)$$

$$\begin{aligned}\frac{dy}{dx} + y^2 &= (e^x - x)' + (e^x - x)^2 \\ &= e^x - 1 + e^{2x} - 2e^x x + x^2 \\ &= e^{2x} + (1 - 2x)e^x + x^2 - 1\end{aligned}$$

(c) Show that $\phi(x) = x^2 - x^{-1}$ is an explicit solution to $x^2 \frac{d^2 y}{dx^2} = 2y$ on the interval $(0, \infty)$.

Answer.

$$\phi(x) = x^2 - x^{-1}, \quad x \in (0, \infty)$$

$$\begin{aligned}x^2 \frac{d^2 y}{dx^2} &= x^2 \cdot (x^2 - x^{-1})'' = x^2 \cdot (2x + x^{-2})' \\ &= x^2 \cdot (2 - 2x^{-3}) \\ &= 2(x^2 - x^{-1}) \\ &= 2\phi(x) = 2y\end{aligned}$$

1.2 (Continued)

3. $y = \sin x + x^2, \quad \frac{d^2 y}{dx^2} + y = x^2 + 2.$

Answer.

$$y = \sin x + x^2$$

$$y' = \cos x + 2x$$

$$y'' = -\sin x + 2$$

$$\frac{d^2y}{dx^2} + y = -\sin x + 2 + \sin x + x^2 = x^2 + 2$$

8. $y = 3 \sin 2x + e^{-x}$, $y'' + 4y = 5e^{-x}$.

Answer.

$$y = 3 \sin 2x + e^{-x}$$

$$y' = 6 \cos 2x - e^{-x}$$

$$y'' = -12 \sin 2x + e^{-x}$$

$$\begin{aligned} y'' + 4y &= -12 \sin 2x + e^{-x} + 4(3 \sin 2x + e^{-x}) \\ &= -12 \sin 2x + e^{-x} + 12 \sin 2x + 4e^{-x} \\ &= 5e^{-x} \end{aligned}$$

10. $y - \ln y = x^2 + 1$, $\frac{dy}{dx} = \frac{2xy}{y-1}$.

Answer.

$$y - \ln y = x^2 + 1$$

$$dy - \frac{1}{y} dy = 2x dx$$

$$\frac{dy}{dx} - \frac{1}{y} \frac{dy}{dx} = 2x$$

$$\left(1 - \frac{1}{y}\right) \frac{dy}{dx} = 2x$$

$$\frac{dy}{dx} = \frac{2x}{1 - \frac{1}{y}} = \frac{2x}{\frac{y-1}{y}} = \frac{2xy}{y-1}$$

11. $e^{xy} + y = x - 1$, $\frac{dy}{dx} = \frac{e^{-xy} - y}{e^{-xy} + x}$.

Answer.

$$e^{xy} + y = x - 1$$

$$d(e^{xy}) + dy = dx$$

$$e^{xy}(x dy + y dx) + dy = dx$$

$$e^{xy} \cdot x \cdot \frac{dy}{dx} + e^{xy} \cdot y + \frac{dy}{dx} = 1$$

$$\frac{dy}{dx}(1 + e^{xy} \cdot x) = 1 - e^{xy} \cdot y$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1 - e^{xy} \cdot y}{1 + e^{xy} \cdot x} = \frac{e^{xy}(e^{-xy} - y)}{e^{xy}(e^{-xy} + x)} \\ &= \frac{e^{-xy} - y}{e^{-xy} + x}\end{aligned}$$

20. Determine for which values of m the function $\phi = e^{mx}$ is a solution to the given equation.

$$(a) \frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 5y = 0,$$

$$(b) \frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} = 0.$$

Answer. (a) $\phi(x) = e^{mx}$

$$\frac{dy}{dx} = m e^{mx}, \quad \frac{d^2 y}{dx^2} = m^2 e^{mx}$$

$$\begin{aligned}\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 5y &= m^2 e^{mx} + 6m e^{mx} + 5e^{mx} \\ &= m^2 e^{mx} + (6m + 5)e^{mx} = 0\end{aligned}$$

$$m^2 + 6m + 5 = 0$$

$$m = \frac{-6 \pm \sqrt{36 - 4}}{2} = -3 \pm 2\sqrt{2}$$

$$(b) \frac{d^3 y}{dx^3} = m^3 e^{mx}$$

$$\frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} = 0$$

$$m^3 e^{mx} + 3m^2 e^{mx} + 2m e^{mx} = 0$$

$$m^3 + 3m^2 + 2m = 0$$

① If $m = 0$: $m^3 + 3m^2 + 2m = 0^3 + 3 \cdot 0^2 + 2 \cdot 0 = 0$

$m = 0$ is a solution.

② If $m \neq 0$:

$$m^2 + 3m + 2 = 0$$

$$(m + 2)(m + 1) = 0$$

$$m = -2 \text{ or } -1$$

Thus, $m = 0$ or -2 or -1 .

22. Verify that the function $\phi(x) = c_1 e^x + c_2 e^{-2x}$ is a solution to the linear equation

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 2y = 0$$

for any choice of the constants c_1 and c_2 . Determine c_1 and c_2 so that each of the following initial conditions are satisfied.

(a) $y(0) = 2, y'(0) = 1$;

(b) $y(1) = 1, y'(1) = 0$.

Answer.

$$\phi(x) = c_1 e^x + c_2 e^{-2x}$$

$$\frac{dy}{dx} = c_1 e^x - 2c_2 e^{-2x}$$

$$\frac{d^2 y}{dx^2} = c_1 e^x + 4c_2 e^{-2x}$$

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 2y = c_1 e^x + 4c_2 e^{-2x} + c_1 e^x - 2c_2 e^{-2x} - 2(c_1 e^x + c_2 e^{-2x})$$

$$= (c_1 + c_1 - 2c_1)e^x + (4c_2 - 2c_2 - 2c_2)e^{-2x} = 0$$

So $\phi(x) = c_1 e^x + c_2 e^{-2x}$ is a solution, $c_1, c_2 \in \mathbb{R}$.

(a)

$$y(0) = c_1 e^0 + c_2 e^0 = c_1 + c_2 = 2$$

$$y'(0) = c_1 e^0 - 2c_2 e^0 = c_1 - 2c_2 = 1$$

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & -2 & 1 \end{array} \right) \xrightarrow{\textcircled{2} + \textcircled{1} \cdot (-1)} \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -3 & -1 \end{array} \right) \quad \begin{cases} c_1 = \frac{5}{3} \\ c_2 = \frac{1}{3} \end{cases}$$

(b)

$$y(1) = c_1 e^1 + c_2 e^{-2} = 1$$

$$y'(1) = c_1 e^1 - 2c_2 e^{-2} = 0$$

$$\left(\begin{array}{cc|c} e & e^{-2} & 1 \\ e & -2e^{-2} & 0 \end{array} \right) \xrightarrow{\textcircled{2} + \textcircled{1} \cdot (-1)} \left(\begin{array}{cc|c} e & e^{-2} & 1 \\ 0 & -3e^{-2} & -1 \end{array} \right) \quad \begin{cases} c_1 = \frac{2}{3e} \\ c_2 = \frac{1}{3}e^2 \end{cases}$$

6.

Use Euler's method with step size $h = 0.2$ to approximate the solution to the initial value problem

$$y' = \frac{1}{x} (y^2 + y), \quad y(1) = 1$$

at the points $x = 1.2, 1.4, 1.6$, and 1.8 .

Answer.

$$y' = f(x, y) = \frac{1}{x} (y^2 + y)$$

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

$$\begin{aligned} y(1.2) &= y(1) + h \cdot f(1, 1) = 1 + 0.2 \times [1 \times (1^2 + 1)] \\ &= 1 + 0.2 \times 2 = 1.4 \end{aligned}$$

$$\begin{aligned} y(1.4) &= y(1.2) + h f(1.2, 1.4) \\ &= 1.4 + 0.2 \times \left[\frac{1}{1.2} \times (1.4^2 + 1.4) \right] \\ &= 1.4 + 0.2 \times 2.8 = 1.96 \end{aligned}$$

$$\begin{aligned}
y(1.6) &= y(1.4) + hf(1.4, 1.96) \\
&= 1.96 + 0.2 \times \left[\frac{1}{1.4} \times (1.96^2 + 1.96) \right] \\
&= 1.96 + 0.2 \times 4.144 = 2.7888
\end{aligned}$$

$$\begin{aligned}
y(1.8) &= y(1.6) + hf(1.6, 2.7888) \\
&= 2.7888 + 0.2 \times \left[\frac{1}{1.6} \times (2.7888^2 + 2.7888) \right] \\
&= 2.7888 + 0.2 \times 6.6039 = 4.1096
\end{aligned}$$

x	y
1.2	1.4
1.4	1.96
1.6	2.7888
1.8	4.1096

Code implementation: [euler.c](#)